

Signed Laplacians for Constrained Graph Clustering

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Motivation & Background



What is clustering?

Problem Definition:

- Group vertices that are close to each other(has many edges between them) into k clusters.
- k is given in the problem.

Standard Clustering:

- Uses only the basic graph structure(vertices and edges) to group vertices

Constrained Clustering:

- Incorporates extra knowledge that are given
- MUST-LINK constraints: vertices that belong in the same cluster(denoted by graph G)
- CANNOT-LINK constraints: vertices that must be separated(denoted by graph H)



Real-world example

Example:

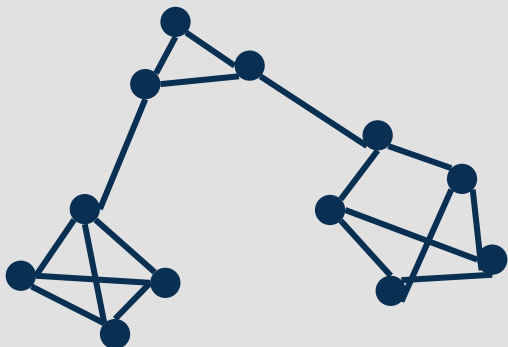
- Grouping weather stations by climate zone.
- Stations A and B always record similar temperatures → MUST-LINK
- Stations C and D are in different climate zones → CANNOT-LINK
- Standard clustering only uses the MUST-LINK data



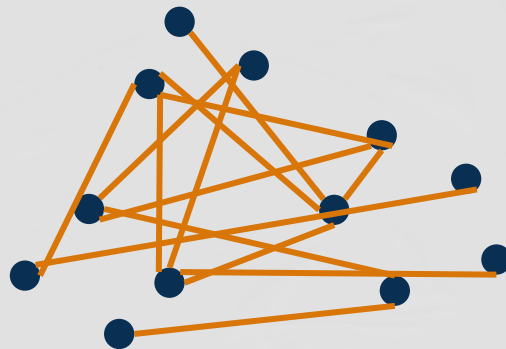
Why not spectral clustering?

Spectral Clustering:

- Uses the second eigenvalue of the Laplacian of the graph
- Looks at only graph G
- Struggles when clusters overlap and need extra information



G(MUST-LINK)



H(CANNOT-LINK)

Background- Spectral Clustering

Spectral Clustering:

- Laplacian $L = D - W$
- Normalized Laplacian $L_{norm} = I - D^{-\frac{1}{2}}WD^{-\frac{1}{2}}$
- We are trying to find a cluster with minimum conductance $\phi(S) = \frac{w(S, V \setminus S)}{vol(S)}$
- Using the second smallest eigenvalue $\lambda_2 = \min_{f \neq 0, f \perp 1} \frac{f^T L f}{f^T f}$
ensures minimum Dirichlet energy $\sum_{u, v \in E} (f(u) - f(v))^2 = f^T L f$
- Low Dirichlet energy means that the differences between neighboring vertices are small, which makes it more likely to select a set with low conductance.
- Cheeger's inequality guarantees that a sweep set algorithm over v_2 produces a cut with conductance $\leq \sqrt{\lambda_2}$



Background- Spectral Clustering

Spectral Clustering:

- Cheeger's inequality

$$\frac{\lambda_2}{2} \leq \phi(G) \leq \sqrt{2\lambda_2}$$

- First part: The best possible conductance cannot be smaller than $\frac{\lambda_2}{2}$
- Second part: There is a set S such that $\phi(S) \leq \sqrt{2\lambda_2}$ and can be found by the sweep set algorithm.
- Sweep set algorithm
Find the Fiedler vector v corresponding to λ_2 .
Sort the entries in the vector so that the entries are nondecreasing. $v_{(1)} \leq v_{(2)} \leq \dots \leq v_{(n)}$
Evaluate the conductances each for selecting the first i vertices ($i = 1, i = 2, \dots, i = n$)
Return the set with lowest conductance



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Experimental results

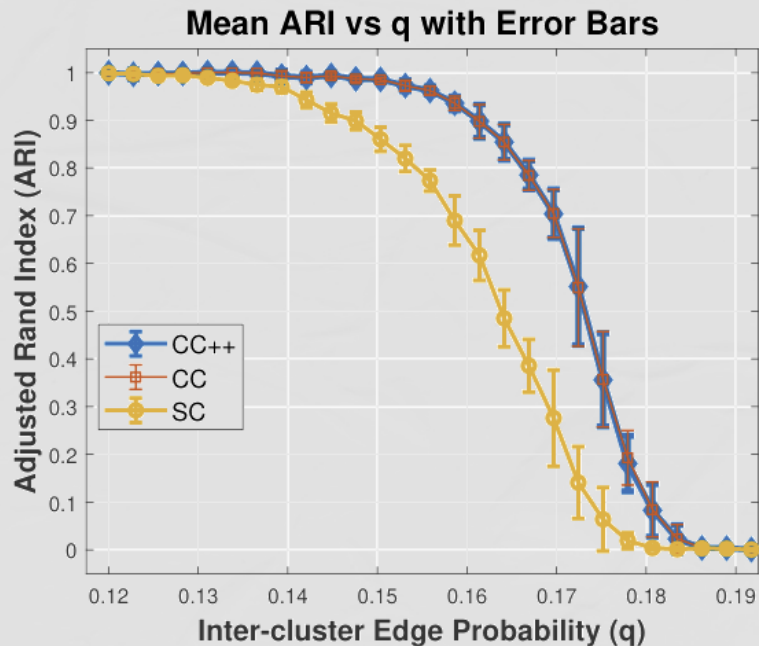


ARI comparison

p: intra-cluster edge probability

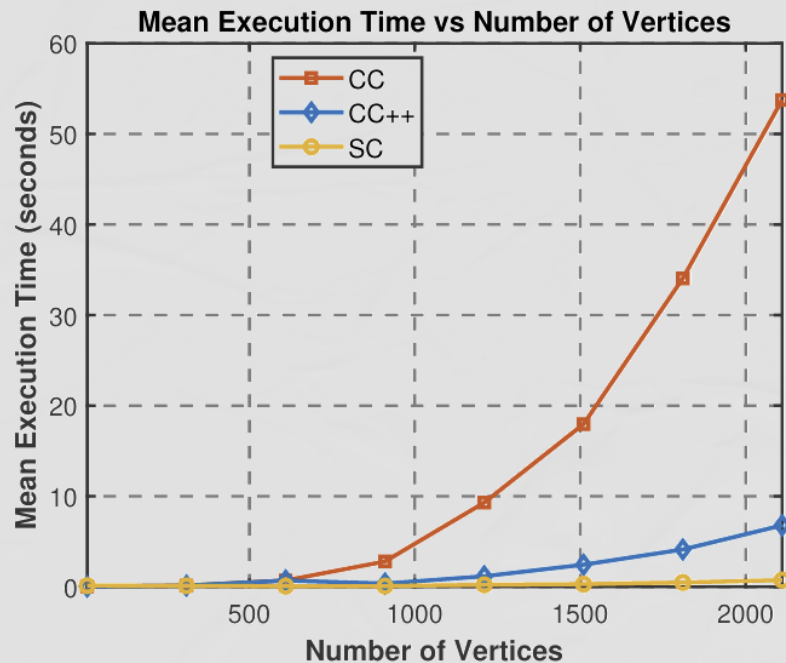
q: inter-cluster edge probability

- Constrained Clustering(CC and CC++) are better than Spectral Clustering(SC) in classifying data according to their known values
- As q increases, distinction between two clusters decrease
- ARI: Adjusted Rand Index. Measures the similarity between two data clusterings or partition



Execution time comparison

- Basic Constrained Clustering(CC) takes much longer than Spectral Clustering(SC).
- The paper's method, Constrained Clustering with loops(CC++) takes much less and CC, being more similar to Spectral Clustering(SC).



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Main algorithm



Main algorithm

1. scale all edge weights so that the $\deg^G(v) = \deg^H(v)$ for all v
2. Compute the normalized Laplacians for G and H
3. Add a small-weighted negative self loop to one vertex in H
4. Solve the generalized eigenvalue problem

$$\frac{\langle f, \Delta^G f \rangle}{\langle f, \Delta^H f \rangle} \text{ subject to } f \perp \mathbb{1}$$

where f is the eigenvector that minimizes the ratio.

5. Apply the sweep-set algorithm on the eigenvector f to partition the vertices of G into two clusters.



Main algorithm

Standard eigenvalue problem

- $L_G x = \lambda x$

Generalized eigenvalue problem

- $L_G x = \lambda L_H x$, where B is symmetric positive definite
- We can convert it to a standard eigenvalue problem if L_H is invertible, by $L_H^{-1} L_G x = \lambda x$
- Usually the Laplacian is singular(not invertible). But adding a negative self-loop makes the matrix invertible
- Basic Constrained Clustering(CC) uses the generalized eigenvalue problem, whereas this paper's algorithm(CC++) uses the standard eigenvalue problem by adding a negative self-loop. The correctness of this method is also proved in the paper.



Main algorithm

Application of the sweep-set algorithm

- The application of the sweep-set algorithm is guaranteed by the Cheeger-type inequality, one of the paper's main results



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Theoretical results



Main result

Cheeger-type inequality for Constrained Clustering

- Let $G = (V, E, w)$ and $H = (V, E', w')$ be graphs such that $\deg^G(v) = \deg^H(v)$ for all $v \in V$.
- Then, it holds that $\phi_H^G \leq 4 \sqrt{\frac{\lambda(\Delta_H^G)}{\lambda_2(\Delta^H)}}$ where Δ^H denotes the normalized Laplacian of the graph H , Δ_H^G is the operator given by $\Delta_H^G(x) = \frac{\langle x, \Delta^G x \rangle}{\langle x, \Delta^H x \rangle}$ and $\lambda_2 = \min_{x \perp 1} \frac{\langle x, \Delta^G x \rangle}{\langle x, \Delta^H x \rangle}$.
- Moreover, the cut achieving this approximation guarantee can be found with a sweep set algorithm.



Main result

Cheeger-type inequality for Constrained Clustering

- We want to prove the existence of a non-empty, proper subset $S \subset V$ such that

$$\frac{w_G(S, V \setminus S)}{w_H(S, V \setminus S)} = 4 \sqrt{\frac{\langle x, \Delta^G x \rangle \cdot \langle x, x \rangle^2}{\langle x, \Delta^H x \rangle}}$$

- To reduce the problem, we find the solution for $z = cx$, where $z_n^w + z_1^2 = 1$.
- Now we want to find a probability density function over $[z_1, z_n]$, and prove

$$\frac{\mathbb{E}(w_G(S, V \setminus S))}{\mathbb{E}(w_H(S, V \setminus S))} = 4 \sqrt{\frac{\langle z, \Delta^G z \rangle \cdot \langle z, z \rangle^2}{\langle z, \Delta^H z \rangle}}$$

- Using the Cauchy-Schwarz inequality, the definition of Δ , and the property $\deg^G(v) = \deg^H(v)$, we can derive the above.
- Define a distribution and choose t according to the probability density function $2|t|$. From this distribution, the bounds of the expectation of the cut ratio matches the intended result.

Main result

Correctness of the idea adding a negative self-loop

- Let $H = (V, E, w)$ be a weighted graph, and let $H' = (V, E', w')$ be another weighted graph such that $E' = E \cup (v_0, v_0)$, where v_0 is a vertex with a self-loop. Assume that $w'|_E = w$ and consider the signature $s = 1$ for all E and $s_{(v_0, v_0)} = -1$. Then, $0 < \lambda_1(\Delta_\alpha^{H'}) \leq \lambda_2(\Delta^H)$, where Δ^H is the Laplacian of graph H .
- Since $\lambda_1(\Delta_\alpha^{H'}) \leq \lambda_2(\Delta^H)$, we can use $\lambda_1(\Delta_\alpha^{H'})$ as the smallest nontrivial eigenvalue



Q&A



Q&A

How does adding a negative self-loop guarantee non-singularity of the Laplacian?

- In the paper, graph signature is denoted as $(\Delta_{\alpha}\varphi)(v) = \frac{1}{deg(v)} \sum_{u \in N_v} (\varphi(v) - \alpha_{vu}\varphi(u)) w_{vu}$.
Here we are using α as an indicator of the sign of the edge, which means that the **weights φ are considered as absolute values**.
- We can interpret this as an extension from $L = D - W$, where for calculating the degrees, absolute values are used.
- Therefore only negative self-loops, and not positive self loops, make the sum $D - W$ nonzero.



Q&A

Can't we just combine the conditions of G and H to one graph and apply SC?

- To answer this question, I referred to “Flexible constrained spectral clustering(Wang, X. and Davidson, I., 2010)”. The paper addresses this issue saying that combining G and H into one graph makes hard, binary edits. Also, it tries to satisfy all given constraints, even if the given constraints are inconsistent.
- In Flexible Constrained clustering, the constraints need to be satisfied by only α while minimizing the cut ratio. α is a bound decided by the algorithm based on the user's setting β .
- In constrained clustering, the objective is to minimize $w^G(S, V \setminus S)$ while maximizing $w^H(S, V \setminus S)$. The **objective is different** from minimizing $w^{G^*}(S, V \setminus S) = w^G(S, V \setminus S) - w^H(S, V \setminus S)$ for example, because this makes negative weights and breaks the Laplacian's property that it is positive semi-definite. If we use some other combinations of G and H, weighting between constraints is necessary but is arbitrary.

Q&A

What is the time complexity of this algorithm?

- The time complexity of CC++ is **polynomial**, just like the time complexity of SC.
- CC++
 - preprocessing: $O(n + m_G)$, where $m_G = |E(G)|$
 - building Laplacians: $O(m_G + m_H)$
 - generalized eigenvalue problem: $O(n^3)$ for dense graphs, $O((m_G + m_H)t)$ for sparse graphs, where m_G, m_H are the number of edges each for G, H and t is the iterations required for convergence
 - sweep set algorithm: $O(n \log n)$ for sorting n vertices, trying n thresholds
- SC
 - building Laplacian: $O(m_G)$
 - standard eigenvalue problem: $O(n^3)$ for dense graphs, $O(m_G t)$ for sparse graphs
 - sweep set algorithm: $O(n \log n)$



Q&A

What is the time complexity of this algorithm?

- Time complexity comparison of solving the eigenvalue problem

	Spectral Clustering	Baseline CC	CC++
Dense	$O(n^3)$ using QR	$O(n^3)$ using QZ, large constant	$O(n^3)$ using Cholesky, small constant
Sparse	$O(m_G t)$ using Lanczos	Cannot use Lanczos (Laplacian singular)	$O((m_G + m_H)t)$ using Lanczos



References

- Carrasco, J. and Sun, H. Signed laplacians for constrained graph clustering. *International Conference on Machine Learning*, 2025.
- Wang, X. and Davidson, I. Flexible constrained spectral clustering. *Proceedings of the 16th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 563–572, 2010.